

Abstract geometric lines in the top-left corner of the slide, consisting of several white lines of varying lengths and orientations that intersect to form a complex, non-representational pattern.

AN IMAGE IS WORTH 16X16 WORDS: TRANSFORMERS FOR IMAGE RECOGNITION AT SCALE

Dept. Artificial Intelligence and MachineLearning

By: Jayanth (4NM22AI400)

Mentor/Guide: Ms. Disha D N

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INTRODUCTION

- Traditional CNNs have been the standard for image processing for many years.
- They analyze images in small parts but struggle to capture long-range connections.
- This limits their ability to understand the full context of an image.
- Transformers, widely used in NLP, handle long-range relationships effectively.
- Researchers introduced Vision Transformers (ViTs) to apply Transformers to images.
- ViTs process images differently and perform better than CNNs on large datasets.
- This marks a shift from traditional CNNs to Transformer-based vision models

SCOPE OF CURRENT TRENDS

- Growing Use of Vision Transformers (ViTs)
- Replacing CNNs
- Better Performance on Large Datasets
- Use in Different Applications
- Advancements in Hardware
- Self-Supervised Learning
- Hybrid Models

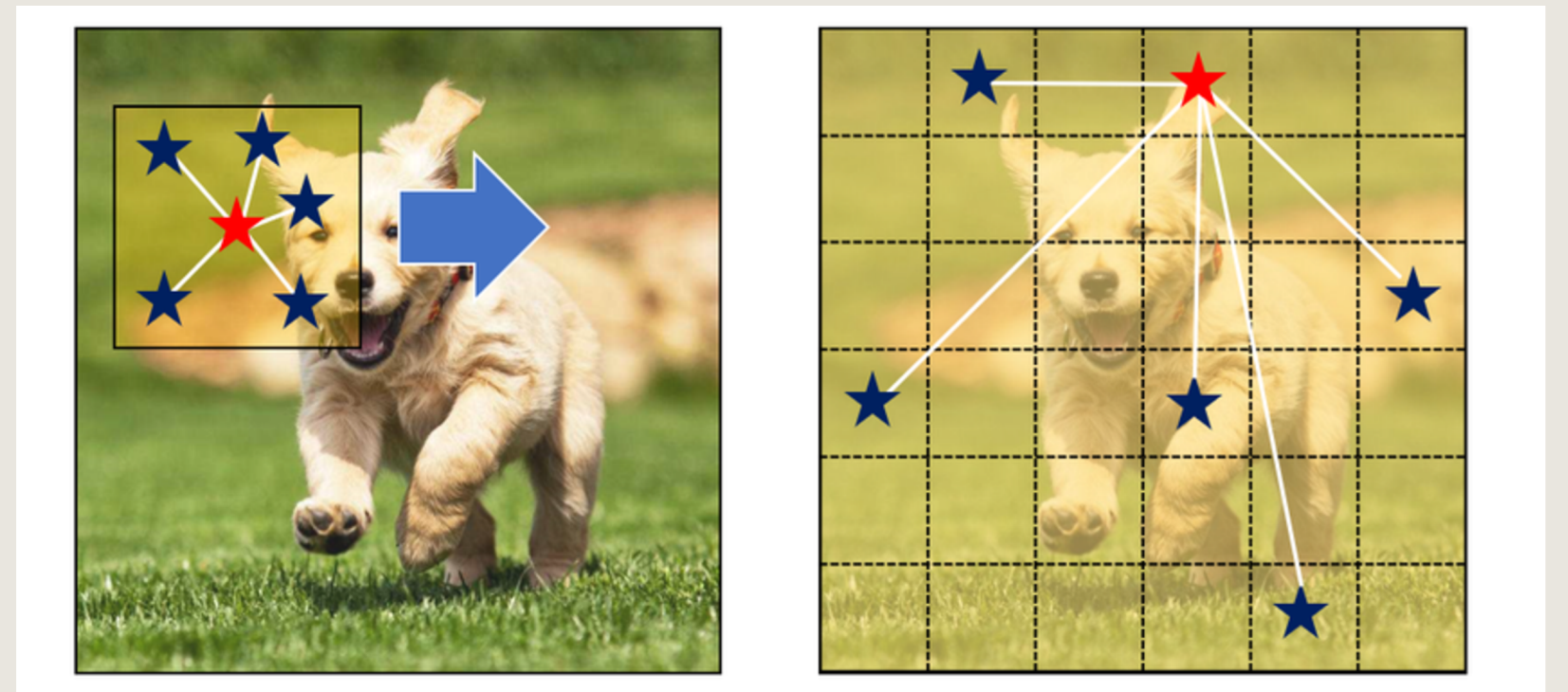
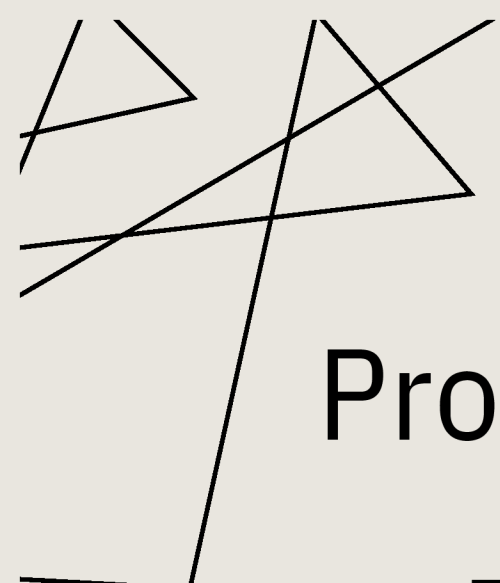
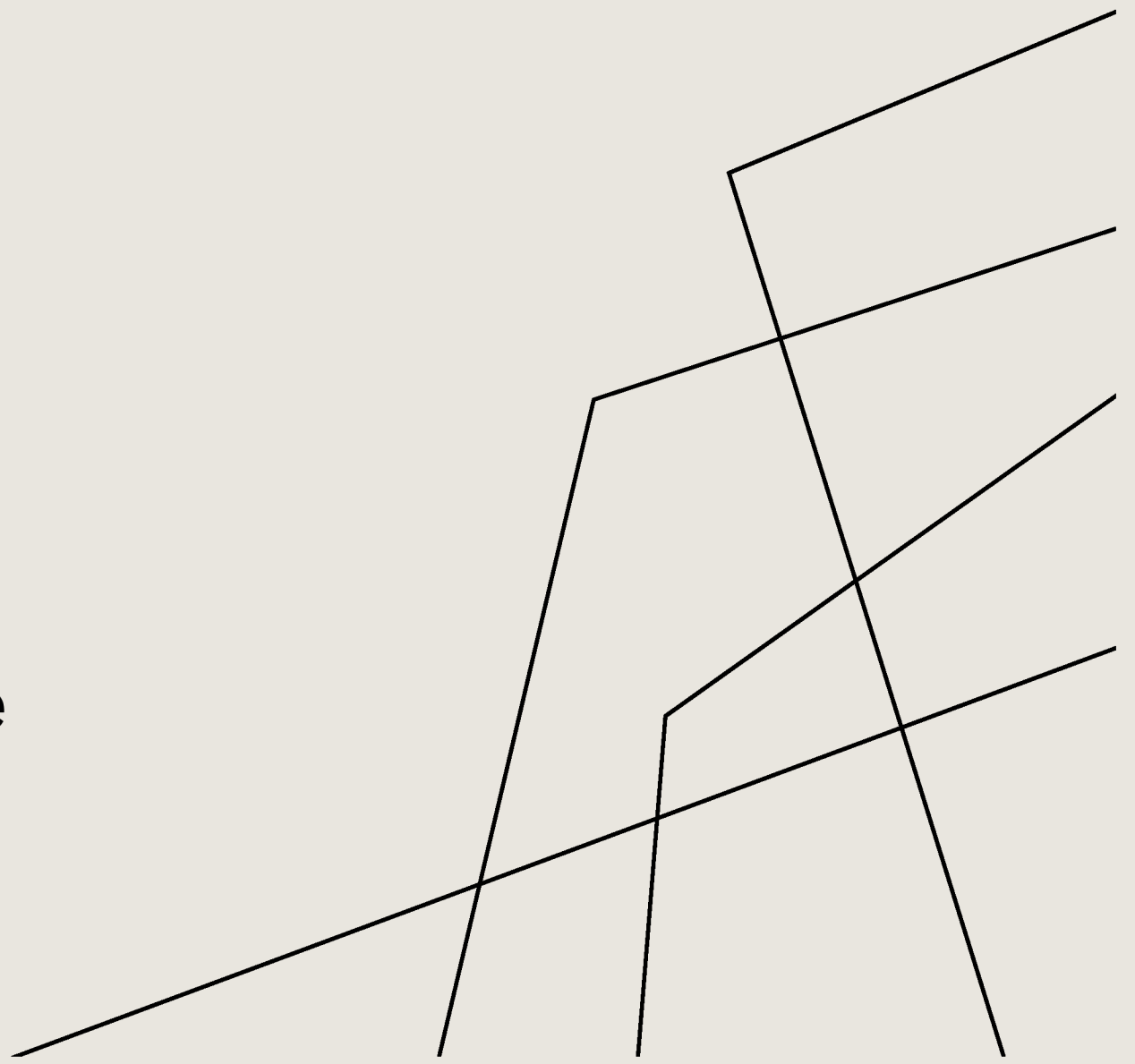


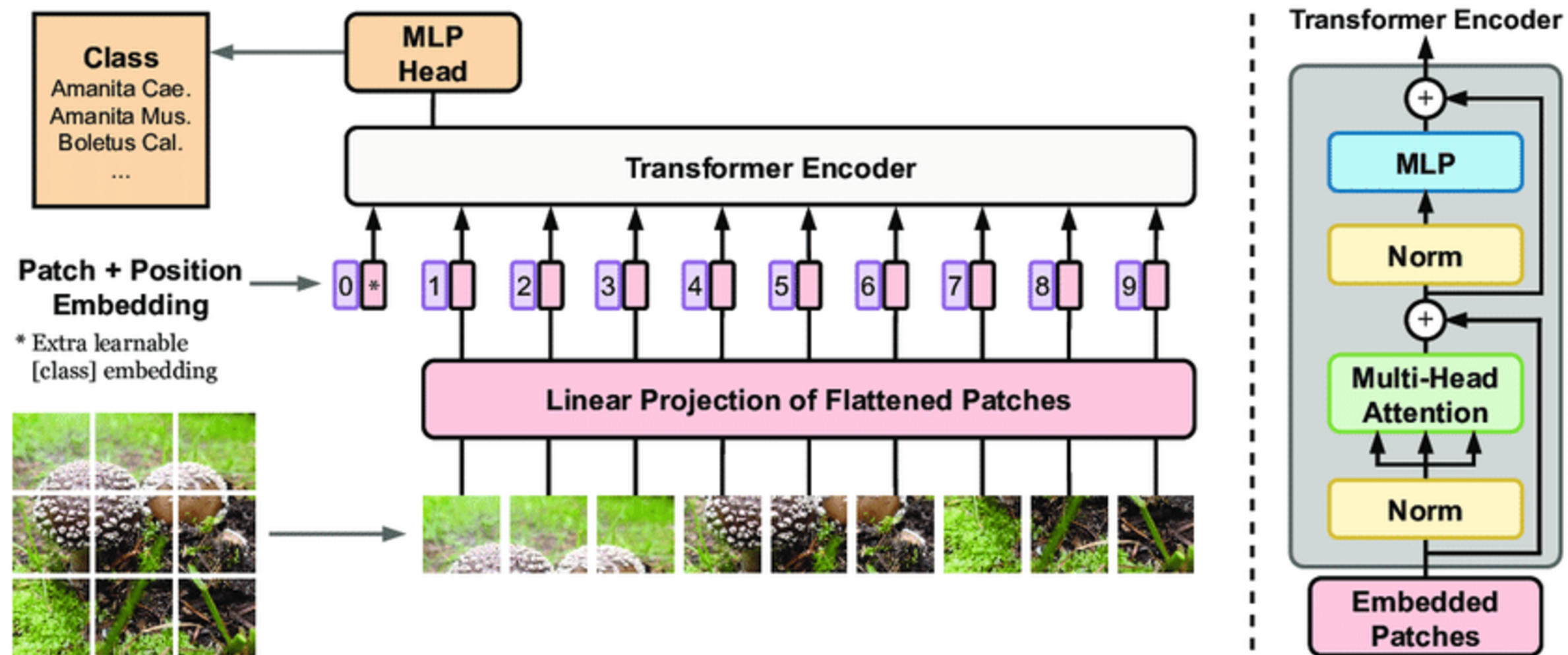
Fig 1: CNN vs Vision Transformer



Proposed Methodology

- Traditional Approach (CNNs)
 - CNNs as the Standard for Image Recognition
 - Pattern Detection with Convolutional Layers
 - Limitations: Struggles with Long-Range Dependencies & Data Requirements
 - New Approach: Vision Transformers (ViTs)
 - ViTs Use Self-Attention Instead of Convolutions
 - Images Are Split into Small Patches Like Words in NLP
 - Patches Are Converted into Embeddings for Processing
 - Transformer Encoder Analyzes Relationships Across the Image
 - Final Output Classifies the Image Like CNNs but Differently
- 

WORKING OF THE TECHNOLOGY – Architecture



- `pip install transformers timm torch torchvision`
- `ViTForImageClassification.from_pretrained("google/vit-base-patch16-224")`
- `ViTFeatureExtractor.from_pretrained("google/vit-base-patch16-224")`

WORKING OF THE TECHNOLOGY – Equations

1 Patch Embedding Equation

- $z_0 = [x_{\text{class}}; x_{p1E}; x_{p2E}; \dots; x_{pNE}] + E_{\text{pos}}$
- $x_{\text{class}} \rightarrow$ Class token | $x_p^i \rightarrow$ Flattened patch | $E \rightarrow$ Embedding matrix | $N \rightarrow$ Number of patches | $E_{\text{pos}} \rightarrow$ Positional embedding | $z_0 \rightarrow$ Embedded patch sequence

2 Transformer Layer

- $Z' = \text{MSA}(\text{LayerNorm}(Z)) + Z$
- $Z = \text{MLP}(\text{LayerNorm}(Z')) + Z'$
- $Z \rightarrow$ Input representation | $\text{LayerNorm}(Z) \rightarrow$ Normalization | $\text{MSA}(Z) \rightarrow$ Multi-Head Self-Attention output | $+Z \rightarrow$ Residual connection | $\text{MLP} \rightarrow$ Multi-Layer Perceptron

3 Final Image Classification Output Equation

- $y = \text{LayerNorm}(z_{L0})$
- $y \rightarrow$ Final output (predicted class) | $z_{L0} \rightarrow$ Class token from last Transformer layer | $\text{LayerNorm} \rightarrow$ Normalization for stability

RESULTS

Dataset	Description	ViT Performance
ImageNet	Large-scale image classification (1,000 classes)	88.55%
ImageNet-Real	Improved ImageNet with better human-verified labels	90.72%
CIFAR-100	Small dataset with 100 object categories	94.55%
VTAB (19 tasks)	19 different image recognition tasks	77.63%
Oxford 102 Flowers	Classifies 102 types of flowers	High Accuracy
Stanford Cars	Recognizes different car models	Strong Performance

KEY HIGHLIGHTS

- ✓ Vision Transformers (ViTs) outperform CNNs when trained on large datasets like ImageNet-21k and JFT-300M.
- ✓ Self-Attention helps ViTs capture long-range dependencies, unlike CNNs which focus on local features.
- ✓ ViTs require less computational power than very deep CNNs like ResNet-152 or EfficientNet.
- ✓ Attention maps show that ViTs focus on key image regions, proving they understand the full context.
- ✓ ViTs perform well across various domains, including object recognition, medical imaging, and autonomous driving.

CONCLUSION

- ◆ ViTs offer a new image processing approach, replacing traditional CNNs.
- ◆ Self-attention helps ViTs understand entire images, unlike CNNs, which focus on small regions.
- ◆ ViTs excel on large datasets but need pretraining to match CNNs on smaller ones.
- ◆ Larger datasets and models improve ViT performance, making them promising for vision tasks.
- ◆ Challenges remain, including high data needs and inefficiency on small datasets.
- ◆ Future models may combine CNNs and Transformers for better efficiency and accuracy.



REFERENCES

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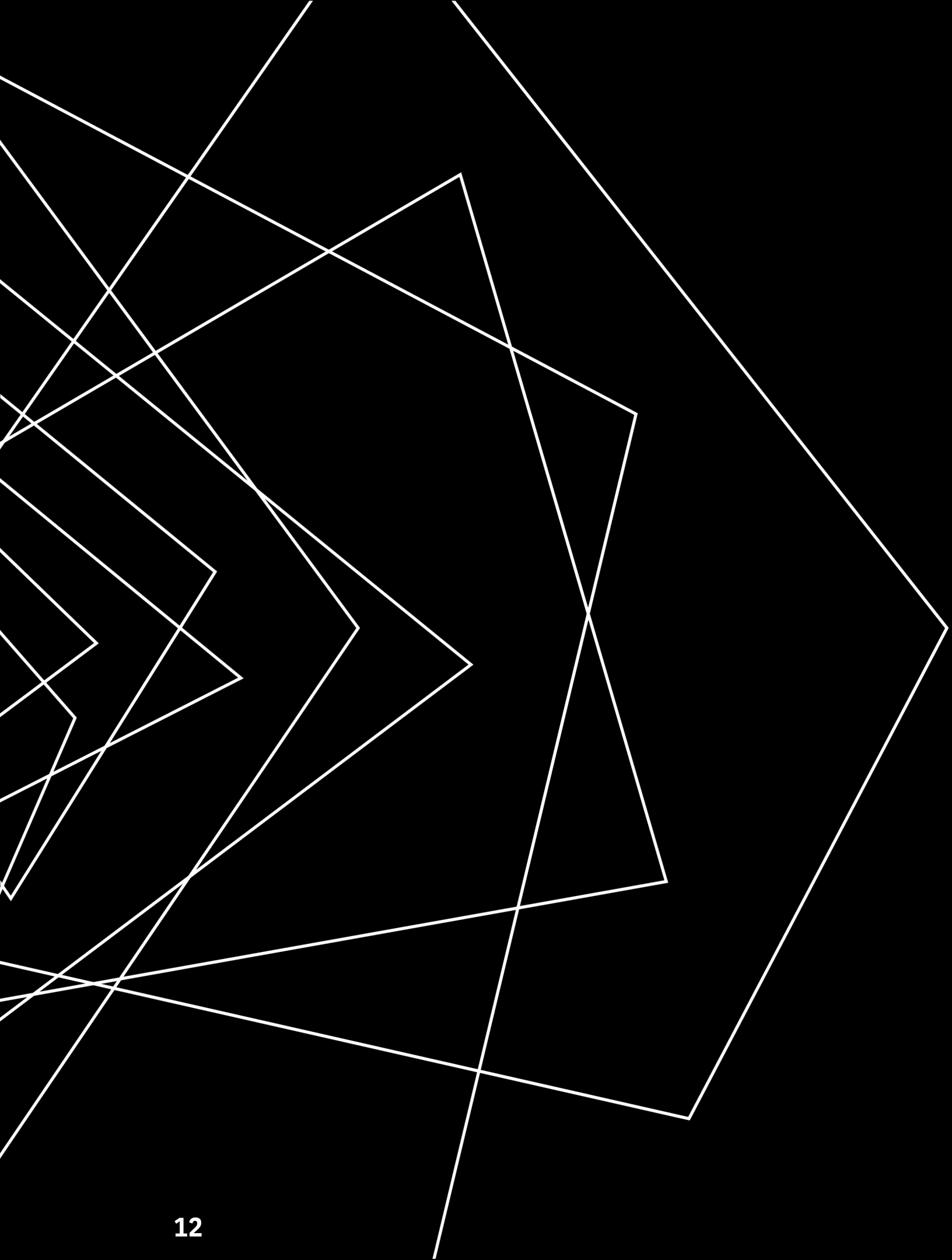
 <https://huggingface.co/spaces/Vokturz/can-it-run-llm>

2 Google AI Blog - Vision Transformer (ViT)

 <https://ai.googleblog.com/2020/12/transformers-for-image-recognition-at.html>

3 Papers with Code - Vision Transformer Leaderboard

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THANK YOU

Jayanth
4NM22AI400